

Single-Pass Estimation of the Clustering Coefficient Distribution in Graph Streams: Supplementary Material

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The Supplementary Material is organized as follows. In Section 1, we provide helpful tools used for proving our results. Section 2 details pseudocodes of subroutines missing from the main paper. Then, Section 3 focuses on proving all our derived results, and Section 4 provides theoretical justification to the choice of some of our parameters. Finally, in Section 5 we provide additional experiments.

1 TOOLS

In the following, we present concentration inequalities used to derive our results.

LEMMA 1.1 (MULTIPLICATIVE CHERNOFF BOUND). *Let $X_1, \dots, X_n$ be independent random variables taking values in $[0, 1]$. Let $X = \sum_{i=1}^n X_i$, and let $\mu = \mathbb{E}[X]$. Then, for any $\delta \in [0, 1]$,*

$$\mathbb{P}[|X - \mu| \geq \delta\mu] \leq 2 \exp\left(-\frac{\delta^2\mu}{3}\right).$$

LEMMA 1.2 (HOEFFDING'S INEQUALITY). *Let $X_1, \dots, X_n$ be random variables with $X_i \in [0, 1]$ that are either independent or negatively associated. Let $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, and $\mu = \mathbb{E}[\bar{X}]$. Then, for any $t \geq 0$,*

$$\mathbb{P}[|\bar{X} - \mu| \geq t] \leq 2 \exp\left(-2nt^2\right)$$

LEMMA 1.3 (BERNSTEIN'S INEQUALITY). *Let $X_1, \dots, X_n$ be independent random variables such that $|X_i - \mathbb{E}[X_i]| \leq M$ for all i . Let $S = \sum_{i=1}^n X_i$, and $V = \sum_{i=1}^n \text{Var}[X_i]$. Then, for any $t \geq 0$,*

$$\mathbb{P}[|S - \mathbb{E}[S]| \geq t] \leq 2 \exp\left(-\frac{t^2}{2V + \frac{2}{3}Mt}\right)$$

LEMMA 1.4 (ONE SIDED CHERNOFF LIKE BOUND). *Let $X_1, \dots, X_n$ be independent random variables such that each $X_i \sim \text{Bern}(p)$ with $p < \tau(1 - \epsilon)$ for some $\epsilon \in (0, 1/2]$. Let $X = \sum_{i=1}^n X_i$. Then, we have:*

$$\mathbb{P}[X \geq n\tau] \leq \exp\left(-\frac{\epsilon^2 n\tau}{10}\right).$$

PROOF. The proof follows immediately from Lemma 22 of [1], where instead of showing concentration on the average $\frac{1}{n} \sum_{i=1}^n X_i$ (as done in [1]), we consider the sum $X = \sum_{i=1}^n X_i$. $\square$

In the following, we leverage a result from [4] in the context of dependency graphs. For random variables $X_1, \dots, X_n$, a graph H

with vertex set $V = \{1, \dots, n\}$ is called a *dependency graph* [3] for variables X_i 's if, for any two disjoint subsets $A, B \subset V$ such that there are no edges connecting a vertex in A to a vertex in B , the collections $\{X_i\}_{i \in A}$ and $\{X_j\}_{j \in B}$ are mutually independent.

LEMMA 1.5 (BERNSTEIN-TYPE INEQ. FOR DEPENDENCY GRAPHS [4, THM. 2.3]). *Let $X_1, \dots, X_n$ be random variables with a dependency graph H and $\mathbb{E}[X_i] = 0$ for all i . Assume $|X_i| \leq b$ for all i , and let $V = \sum_{i=1}^n \text{Var}[X_i]$. Let $\chi^*(H)$ be the fractional chromatic number of H . Then for all $t \geq 0$,*

$$\mathbb{P}\left[\left|\sum_{i=1}^n X_i\right| \geq t\right] \leq 2 \exp\left(-\frac{8t^2}{25\chi^*(H)\left(V + \frac{bt}{3}\right)}\right).$$

Above, the fractional chromatic number $\chi^*(H)$ is defined as the linear programming relaxation of the standard chromatic number $\chi(H)$. Intuitively, it represents the minimum weight needed to cover the vertices of H with independent sets. It holds $\chi^*(H) \leq \chi(H) \leq d_{\max}(H) + 1$, where $d_{\max}(H)$ is the maximum degree for graph H .

2 ADDITIONAL ALGORITHMS

In this section, we report the pseudocode of subroutines missing from our main paper.

SampleEdge (Algorithm 1) adds the given edge independently to the given sample with the given probability.

CTReservoir (Algorithm 2) is given the incoming edge $\{u, v\}$, the main sample S with its memory budget M_s and counter s , the auxiliary sample A with its budget M_a and counter a , and the node-triangle map to eventually update. First, the algorithm computes the adjacency lists of nodes u, v in both samples S and A , deriving the list of neighbors (lines 1-4). Then, for every node w forming a triangle with u and v (line 5), it derives the probability of sampling such triangle based on whether the two edges (when the incoming one arrives) forming the triangle belong both to sample S (line 8), both to sample A (line 10), or one in S and the other in A (line 11). Finally, such probability is used to increment the local triangles count (line 12).

SEReservoir (Algorithm 3) is given the edge to sample, the reservoir sample S together with its memory budget M and the number s of elements seen so far (belonging to S), and applies the reservoir sampling update: if S is not full yet, it adds the incoming edge to S (line 1), otherwise it adds it to S with probability M/s and by replacing an edge in S uniformly at random (lines 2-4). Then, it increments the counter s (line 5). We assume that **FlipBiasedCoin**(p) returns **HEAD** with probability p .

Notice that the correctness of our triangle-counter algorithm follows from the use of reservoir sampling (see [9, 10]), and since

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Algorithm 1: SampleEdge($\{u, v\}, S, p$)

Input: edge $\{u, v\}$; edge sample S ; probability p of sampling;
1 Let $S \leftarrow S \cup \{\{u, v\}\}$ with probability p ;

Algorithm 2: CTReservoir($\{u, v\}, S, M_s, s, A, M_a, a, \hat{T}$)

Input : edge $\{u, v\}$;
main sample S ; memory budget M_s for sample S ;
number s of edges belonging to S ;
auxiliary sample A ; memory budget M_a for sample A ;
number a of edges belonging to A ;
Node-triangle map $\hat{T}$.
1 Let $Adj_S(u), Adj_S(v)$ be the adjacency lists of u, v in sample S ;
2 Let $Adj_A(u), Adj_A(v)$ be the adjacency lists of u, v in sample A ;
3 Let $N(u) \leftarrow Adj_S(u) \cup Adj_A(u)$;
4 Let $N(v) \leftarrow Adj_S(v) \cup Adj_A(v)$;
5 **for** each node $w \in N(u) \cap N(v)$ **do**
6 Initialize $\hat{T}[u], \hat{T}[v], \hat{T}[w]$ to zero if not set yet;
7 **if** $w \in Adj_S(u)$ AND $w \in Adj_S(v)$ **then**
8 $p \leftarrow \min\left(1, \frac{M_s}{s} \cdot \frac{M_s-1}{s-1}\right)$;
9 **else if** $w \in Adj_A(u)$ AND $w \in Adj_A(v)$ **then**
10 $p \leftarrow \min\left(1, \frac{M_a}{a} \cdot \frac{M_a-1}{a-1}\right)$;
11 **else** $p \leftarrow \min\left(1, \frac{M_s}{s} \cdot \frac{M_a}{a}\right)$;
12 Increment $\hat{T}[u], \hat{T}[v], \hat{T}[w]$ by p ;

Algorithm 3: SEReservoir($\{u, v\}, S, M, s$)

Input : edge $\{u, v\}$; edge sample S ;
memory budget M for sample S ;
number s of edges belonging to S seen so far.
1 **if** $|S| < M$ **then** $S \leftarrow S \cup \{\{u, v\}\}$;
2 **else if** FlipBiasedCoin(M/s) == HEAD **then**
3 $\{\bar{u}, \bar{v}\} \leftarrow$ edge sampled uniformly at random from S ;
4 $S \leftarrow (S \setminus \{\{\bar{u}, \bar{v}\}\}) \cup \{\{u, v\}\}$;
5 $s \leftarrow s + 1$;

we are using independent reservoir samples for (head and tail) main and auxiliary samples (see Section 5 of the main paper).

3 ADDITIONAL PROOFS

We report all our derived results together with their proofs.

LEMMA 3.1. Let $\hat{d}_u$ be the estimated degree for each node u contained in node-degree maps S_h and S_t returned by FirstPass algorithm, using fixed probability p_t for sampling edges. Then:

(i) $\hat{d}_u = d_u$ for each node $u \in S_h$,

and, for any $\varepsilon \in (0, 1/2)$, if $\tau = \tilde{\Omega}\left(\frac{1}{\varepsilon p_t}\right)$, we have:

(ii) if $d_v \leq (1 - \varepsilon)\tau$, then $\hat{d}_v \leq \tau$ for each node $v \in S_t$;

(iii) if $d_v \geq \tau$, then $\hat{d}_v \in (1 \pm \varepsilon)d_v$ for each node $v \in S_t$ with high probability;

(iv) if $d_v \geq \tau$, then $v \in S_t$ with high probability.

PROOF. *Item (i).* By design, FirstPass is tracking exactly degrees of nodes in S_h , which are sampled via hashing.

Item (ii). Fix a vertex v . Let C_v be the “raw” counter for node v , i.e., the value in $S_t[v]$ before adding back the expected loss. We can express $d_v = C_v + L_v$, where L_v is the number of edges incident to v that we missed. For any $a \geq 0$, we have $\mathbb{P}[L_v \geq a] \leq (1 - p_t)^a \leq \exp(-p_t a)$. The expected loss $\ell(r)$ is such that $0 \leq \ell(r) \leq \lceil \frac{1-p_t}{p_t} \rceil \leq \frac{1}{p_t}$. For $d_v \leq (1 - \varepsilon)\tau$ and the suggested choice of τ :

$$\hat{d}_v = C_v + \ell\left(\hat{d}_v\right) \leq d_v + \frac{1}{p_t} \leq (1 - \varepsilon)\tau + \varepsilon\tau \leq \tau.$$

Item (iii). Fix a vertex v . For the lower tail, we have that if $L_v \leq \varepsilon d_v$, which happens with probability $1 - \exp(-p_t \varepsilon d_v) \geq 1 - \exp(-p_t \varepsilon \tau)$, then:

$$\hat{d}_v = C_v + \ell\left(\hat{d}_v\right) \geq C_v = d_v - L_v \geq d_v - \varepsilon d_v = (1 - \varepsilon)d_v.$$

Therefore, the lower tail follows by a union bound and the suggested choice of τ . The upper tail holds deterministically:

$$\hat{d}_v = C_v + \ell\left(\hat{d}_v\right) \leq d_v + \frac{1}{p_t} \leq d_v + \varepsilon\tau \leq (1 + \varepsilon)d_v.$$

Item (iv). Fix a vertex v such that $d_v \geq \tau$. Then:

$$\mathbb{P}[v \notin S_t] = \mathbb{P}[L_v \geq d_v] \leq \exp(-p_t d_v) \leq \exp(-p_t \tau) \leq \frac{1}{nc}.$$

The item follows by a union bound over nodes v with $d_v \geq \tau$. $\square$

FACT 3.2. Given $\hat{d}$ such that $\hat{d} \in (1 \pm \varepsilon)d$, and $d \geq \tau$, it holds:

$$\binom{\hat{d}}{2} \in \left[(1 - \varepsilon)^2 - \frac{\varepsilon(1 - \varepsilon)}{\tau - 1}, (1 + \varepsilon)^2 + \frac{\varepsilon(1 + \varepsilon)}{\tau - 1} \right] \binom{d}{2}.$$

COROLLARY 3.3. Following Fact 3.2, for any $\varepsilon \in (0, 1/2]$, $\tau \geq 4$:

$$\binom{\hat{d}}{2} \in (1 \pm 3\varepsilon) \binom{d}{2}.$$

PROOF. From Fact 3.2, we analyze the upper and lower bounds, separately. For the upper bound, since $\varepsilon \leq 1/2$, we have $\varepsilon^2 \leq \varepsilon/2$. Thus:

$$(1 + \varepsilon)^2 + \frac{\varepsilon(1 + \varepsilon)}{\tau - 1} = 1 + 2\varepsilon + \varepsilon^2 + \frac{\varepsilon(1 + \varepsilon)}{\tau - 1} \leq 1 + 2.5\varepsilon + \frac{1.5\varepsilon}{\tau - 1}.$$

For $\tau \geq 4$, it follows an upper bound of $1 + 3\varepsilon$. For the lower bound:

$$(1 - \varepsilon)^2 - \frac{\varepsilon(1 - \varepsilon)}{\tau - 1} = 1 - 2\varepsilon + \varepsilon^2 - \frac{\varepsilon(1 - \varepsilon)}{\tau - 1} \geq 1 - 2\varepsilon - \frac{\varepsilon}{\tau - 1}.$$

For $\tau \geq 4$, the expression is $> 1 - 3\varepsilon$. $\square$

Notice that in our experiments, we have values of τ from 253 (as-skitter) to 4448 (twitter).

Quality of Estimates. In the following, we report proofs of results analyzing the quality of the estimates for binned-degree clustering coefficients, returned by our proposed (two-pass) approach. Note that such results extend to our one-pass approach (see Section 5 of the main paper).

We start by analyzing the head estimator. We prove Theorem 4.1 of main paper by showing bounds for NDCC and WDCC separately in Theorem 3.6 and 3.7. First, we provide concentration for estimates of the number of nodes and the number of triangles. Throughout, we assume degree interval $D = [L, U]$, with $U = \Theta(L)$.

LEMMA 3.4. Consider a degree interval D and any $\varepsilon \in (0, 1)$. Let $C_h(D)$ be the number of nodes with degree in D in the sample S_h . Then, if $p_h = \tilde{\Omega}\left(\frac{1}{\varepsilon^2 C(D)}\right)$, we have, with high probability, $C_h(D) \in (1 \pm \varepsilon)p_h C(D)$.

PROOF. Let $X_u = 1$ if $u \in S_h$ and 0 otherwise. Note that X_i 's are independent. We have $C_h(D) = \sum_{u, d_u \in D} X_u$, and $\mathbb{E}[C_h(D)] = p_h C(D)$. Result follows from an application of Lemma 1.1. $\square$

We show concentration bounds also for the number of triangles incident to sampled nodes. Recall that $\Delta_V(D)$ is the maximum number of triangles incident to a node with degree in D .

LEMMA 3.5. Consider degree interval D and any $\varepsilon \in (0, 1)$. Let $T_h(D)$ be the number of triangles incident to nodes with degree in D in the sample S_h . Then, if $p_h = \tilde{\Omega}\left(\frac{\Delta_V(D)}{\varepsilon^2 T(D)}\right)$, we have, with high probability, $T_h(D) \in (1 \pm \varepsilon)p_h T(D)$.

PROOF. Let $X_u = 1$ if $u \in S_h$ and 0 otherwise. Note that X_i 's are independent. We have $T_h(D) = \sum_{u, d_u \in D} X_u t_u$, and $\mathbb{E}[T_h(D)] = p_h T(D)$. By definition, it holds $X_u t_u \leq \Delta_V(D)$, for each u . Result follows by an application of Lemma 1.1 to the normalized variables $X_u t_u / \Delta_V(D)$. $\square$

THEOREM 3.6. Consider degree interval $D = [L, U)$ such that $L < U \leq \tau$. For any $\varepsilon, \eta \in (0, 1)$, if

$$p_h = \tilde{\Omega}\left(\frac{1}{\eta^2} \left(\frac{1}{C(D)} + \frac{\Delta_V(D)}{T(D)}\right)\right) \text{ and}$$

$$p_{M_h} p_{A_h} = \tilde{\Omega}\left(\frac{1}{\varepsilon^2} \left(\eta^2 + \frac{\Delta_E(D)}{T(D)}\right)\right),$$

then Algorithm Estimate outputs a value $\widehat{NDCC}(D)$ such that:

$$\left| \widehat{NDCC}(D) - NDCC(D) \right| \leq \varepsilon \cdot NDCC(D) + \eta.$$

with high probability.

PROOF. Recall that $\hat{d}_u = d_u$ for any $u \in S_h$, since degrees are tracked exactly (Lemma 3.1, item (i)). For head estimator, Alg. Estimate computes estimates as:

$$\widehat{NDCC}(D) = \frac{1}{|\{u \in S_h, d_u \in D\}|} \sum_{u \in S_h, d_u \in D} \frac{\hat{t}_u}{\binom{d_u}{2}}. \quad (1)$$

Let $\widehat{C}(D) = |\{u \in S_h, d_u \in D\}|$. Define the quantity:

$$Y(D) = \frac{1}{\widehat{C}(D)} \sum_{u \in S_h, d_u \in D} \frac{t_u}{\binom{d_u}{2}}$$

We bound the overall error as follows:

$$\begin{aligned} & \left| \widehat{NDCC}(D) - NDCC(D) \right| \\ & \leq \underbrace{|\widehat{NDCC}(D) - Y(D)|}_{(i): \text{node sampling error}} + \underbrace{|Y(D) - NDCC(D)|}_{(ii): \text{triangle estimator error}}. \end{aligned} \quad (2)$$

First, we bound term (i) of Eq. 2. Let $Z_u = t_u / \binom{d_u}{2}$ if $u \in S_h$, and 0 otherwise; we have $Z_u \in [0, 1]$ and $Y(D) = \sum_{u, d_u \in D} Z_u / \widehat{C}(D)$. Conditioning on the event $\widehat{C}(D) = C \in (1 \pm \eta)p_h C(D)$ (see

Lemma 3.4), we have that variables Z_i 's are negatively associated. That is, $\mathbb{P}[Z_u \neq 0 \mid \widehat{C}(D) = C] = C / C(D)$. Therefore:

$$\begin{aligned} & \mathbb{E}[Y(D) \mid \widehat{C}(D) = C] \\ &= \frac{1}{C} \sum_{u, d_u \in D} \frac{t_u}{\binom{d_u}{2}} \mathbb{E}\left[u \in S_h, d_u \in D \mid \widehat{C}(D) = C\right] \\ &= \frac{1}{C} \sum_{u, d_u \in D} \frac{t_u}{\binom{d_u}{2}} \frac{C}{C(D)} = NDCC(D). \end{aligned}$$

Applying Hoeffding's inequality (Lemma 1.2) for negative associated variables:

$$\begin{aligned} & \mathbb{P}\left[|Y(D) - NDCC(D)| \geq \eta \mid \widehat{C}_h = C\right] \\ &= \mathbb{P}\left[\left|\frac{\sum_{u, d_u \in D} Z_u}{C} - NDCC(D)\right| \geq \eta \mid \widehat{C}_h = C\right] \\ &\leq 2 \exp(-2C\eta^2) \leq 2 \exp(-2(1 - \eta)p_h C(D)\eta^2), \end{aligned}$$

which is $\leq 1/n^c$ for $p_h = \tilde{\Omega}\left(\frac{1}{\eta^2 C(D)}\right)$, which bounds the first term of Eq. 2. Applying a union bound over both events concludes the proof for case 1.

Now, we bound term (ii) of Eq. 2, which controls the error of our triangle-estimator. Our goal boils down to bounding:

$$\frac{1}{\widehat{C}(D)} \left| \sum_{u \in S_h, d_u \in D} \frac{\hat{t}_u - t_u}{\binom{d_u}{2}} \right|. \quad (3)$$

Define $\widehat{A}(D), A(D)$ as $\sum_{u \in S_h, d_u \in D} \frac{\hat{t}_u}{\binom{d_u}{2}}$ and $\sum_{u \in S_h, d_u \in D} \frac{t_u}{\binom{d_u}{2}}$, respectively. Eq. 3 can be expressed as $\frac{1}{\widehat{C}(D)} |\widehat{A}(D) - A(D)|$. Let $\mathcal{T}_h(D)$ be the set of triangles incident to nodes in S_h , having degree in D . We have $|\mathcal{T}_h(D)| = T_h(D)$. For each triangle Δ , define:

$$w_\Delta = \sum_{u \in \Delta \cap S_h, d_u \in D} \frac{1}{\binom{d_u}{2}} \in \left[\frac{1}{\binom{U}{2}}, \frac{3}{\binom{L}{2}} \right].$$

We have $\sum_{\Delta \in \mathcal{T}_h(D)} w_\Delta = A(D)$. Let $I_\Delta = 1$ if triangle Δ is counted by Alg. SecondPass, and 0 otherwise. We can write:

$$\widehat{A}(D) = \sum_{\Delta \in \mathcal{T}_h(D)} \frac{w_\Delta}{p_\Delta} I_\Delta, \text{ and } \mathbb{E}[\widehat{A}(D)] = A(D).$$

Let $X_\Delta = \frac{w_\Delta I_\Delta}{p_\Delta}$. We have $\text{Var}[X_\Delta] \leq \frac{9p_\Delta(1-p_\Delta)}{\binom{L}{2}^2 p_\Delta^2} \leq \frac{9}{\binom{L}{2}^2 p_A p_M}$, assuming $p_A \leq p_M$. Define centered variables $Y_\Delta = X_\Delta - \mathbb{E}[X_\Delta]$. We have $Y_\Delta = w_\Delta \left(\frac{I_\Delta}{p_\Delta} - 1\right) \leq \frac{3}{\binom{L}{2} p_M p_A}$. Note that $\sum_\Delta X_\Delta = \widehat{A}(D)$.

Consider the dependency graph H for variables $X_1, \dots, X_{T_h(D)}$. That is, place a vertex in H for every variable X_i , and place an edge between X_{Δ_i} and X_{Δ_j} if they are dependent, i.e., if triangle Δ_i and triangle Δ_j share a sampled edge during the streaming algorithm.

We provide a refined bound for the fractional chromatic number $\chi^*(H) \leq \chi$, which corresponds to the maximum number of triangles (among those in $\mathcal{T}_h(D)$) sharing an edge in the original graph. Fix an edge $e = \{a, b\}$. We have the following cases:

- if $a \in V_D$ and is sampled in S_h , then every triangle adjacent to edge e is incident to a . Hence, $\chi \leq \Delta_V(D)$. The same holds when $b \in V_D$ and $b \in S_h$.

- if $a, b \notin V_D$, then a triangle incident to e is counted when the third endpoint is in V_D and sampled in S_h . Hence, the number of distinct triangles through e corresponds to the number of nodes (in $S_h \cap V_D$) forming triangles e . Let t_e^{3rd} be the number of triangles adjacent to edge e with the third endpoint in V_D . We express $X_e \sim \text{Bin}(t_e^{3rd}, p_h)$, as each node is sampled i.i.d. via hashing. Let $\mu_e = t_e^{3rd} p_h$. If $\mu_e = \Omega(\log n)$, by a Chernoff bound we get $\mathbb{P}[X_e \geq 2\mu_e] \leq \exp(-\mu_e/3) \leq n^{-c-2}$. Otherwise, if $\mu_e = o(\log n)$, we have: $\mathbb{P}[X_e \geq a] \leq \sum_{k \geq a} \binom{t_e^{3rd}}{k} p_h^k \leq \sum_{k \geq a} \left(\frac{e\mu_e}{k}\right)^k \leq 2 \left(\frac{e\mu_e}{a}\right)^a$ by solving the geometric series for $a \geq 2e\mu_e$. Picking $a = \max(2e\mu_e, O(\log n))$ makes this probability small enough.

Given that $t_e^{3rd} \leq \Delta_E(D)$ and by a union bound over $\Theta(n^2)$ many edges, we get $\chi \leq \Delta_V(D) + p_h \Delta_E(D) + \log n$ (with high probability)

Now, we have $\sum_{\Delta} \text{Var}[Y_{\Delta}] = \sum_{\Delta} \text{Var}[X_{\Delta}] \leq T_h(D) \frac{9}{\binom{L}{2} p_{MPA}}$.

By defining $Y = \sum_{\Delta} Y_{\Delta}$, we can apply Lemma 1.5:

$$\begin{aligned} \mathbb{P}\left[\left|\widehat{A}(D) - A(D)\right| \geq \varepsilon A(D)\right] &= \mathbb{P}\left[|Y| \geq \varepsilon A(D)\right] \\ &\leq 2 \exp\left(\frac{-8\varepsilon^2 A^2(D)}{25\chi \left(\frac{9T_h(D)}{\binom{L}{2} p_{MPA}} + \frac{\varepsilon A(D)}{\binom{L}{2} p_{MPA}}\right)}\right), \end{aligned}$$

which is $\leq 1/n^c$ given that $A(D) \in T_h(D) \left[1/\binom{U}{2}, 3/\binom{L}{2}\right]$, $U/L = \Theta(1)$, for $p_{MPA} = \widetilde{\Omega}\left(\frac{\chi}{\varepsilon^2 T_h(D)}\right)$. We then use Lemma 3.5 to obtain concentration for $T_h(D)$ around $p_h T(D)$. Finally, we use the fact that $1/C(D) \leq \Delta_V(D)/T(D)$ to get $p_{MPA} = \widetilde{\Omega}\left(\frac{1}{\varepsilon^2} \left(\eta^2 + \frac{\Delta_E(D)}{\varepsilon^2 T(D)}\right)\right)$. Thus, we showed:

$$\left|\widehat{NDCC}(D) - Y(D)\right| = \frac{1}{\widehat{C}(D)} \left|\widehat{A}(D) - A(D)\right| \leq \frac{\varepsilon A(D)}{\widehat{C}(D)} = \varepsilon Y(D),$$

with high probability. Combining item (i), (ii) we proved, for $D = [L, U]$ with $L \leq U \leq \tau$, that:

$$\begin{aligned} \left|\widehat{NDCC}(D) - NDCC(D)\right| &\leq \eta + \varepsilon Y(D) \leq \eta + \varepsilon(NDCC(D) + \eta) \\ &= \varepsilon \cdot NDCC(D) + (1 + \varepsilon)\eta. \end{aligned}$$

The final result follows then by rescaling η .

□

THEOREM 3.7. Consider a degree interval $D = [L, U]$ such that $L < U \leq \tau$. For any $\varepsilon, \eta \in (0, 1)$, if

$$\begin{aligned} p_h &= \widetilde{\Omega}\left(\frac{1}{\eta^2} \left(\frac{1}{C(D)} + \frac{\Delta_V(D)}{T(D)}\right)\right) \text{ and} \\ p_{Mh} p_{Ah} &= \widetilde{\Omega}\left(\frac{1}{\varepsilon^2} \left(\eta^2 + \frac{\Delta_E(D)}{T(D)}\right)\right), \end{aligned}$$

then Algorithm Estimate outputs a value $\widehat{WDCC}(D)$ such that:

$$\left|\widehat{WDCC}(D) - WDCC(D)\right| \leq \varepsilon \cdot WDCC(D) + \eta.$$

PROOF. Recall that $\widehat{d}_u = d_u$ for any $u \in S_h$, since degrees are tracked exactly (Lemma 3.1, item (i)). For the head estimator, Alg.

Estimate computes:

$$\widehat{WDCC}(D) = \frac{\sum_{u \in S_h, d_u \in D} \widehat{t}_u}{\sum_{u \in S_h, d_u \in D} \binom{d_u}{2}}. \quad (4)$$

Define the quantity using true triangle counts:

$$Y(D) = \frac{\sum_{u \in S_h, d_u \in D} t_u}{\sum_{u \in S_h, d_u \in D} \binom{d_u}{2}}.$$

We bound the overall error as follows:

$$\begin{aligned} &\left|\widehat{WDCC}(D) - WDCC(D)\right| \\ &\leq \underbrace{\left|WDCC(D) - Y(D)\right|}_{(i): \text{ node sampling error}} + \underbrace{\left|\widehat{WDCC}(D) - Y(D)\right|}_{(ii): \text{ triangle estimator error}}. \end{aligned} \quad (5)$$

First, we bound term (i) of Eq. 5. Let $X_u = 1$ if $u \in S_h$ and 0 otherwise. For the upper tail, we have:

$$\begin{aligned} &\mathbb{P}[Y(D) \geq WDCC(D) + \eta] \\ &= \mathbb{P}\left[\sum_{u, d_u \in D} X_u t_u \geq (WDCC(D) + \eta) \sum_{u, d_u \in D} X_u \binom{d_u}{2}\right] \\ &= \mathbb{P}\left[\sum_{u, d_u \in D} Z_u \geq 0\right], \end{aligned}$$

by defining $Z_u = X_u \left(t_u - (WDCC(D) + \eta) \binom{d_u}{2}\right)$. Note that Z_u 's are independent. Let $Z = \sum_{u, d_u \in D} Z_u$. We have:

$$\begin{aligned} \mathbb{E}[Z] &= p_h \sum_{u, d_u \in D} \left(t_u - WDCC(D) \binom{d_u}{2} - \eta \binom{d_u}{2}\right) \\ &= -\eta \sum_{u, d_u \in D} p_h \binom{d_u}{2} \geq -\eta C(D) p_h \binom{U}{2}, \end{aligned}$$

as $D = [L, U]$. Moreover, each $|Z_u|$ is upper bounded by $\binom{d_u}{2} \leq \binom{U}{2}$, and $\text{Var}[Z] \leq p_h C(D) \binom{U}{2}^2$. By Bernstein's inequality (Lemma 1.3):

$$\mathbb{P}[Z - \mathbb{E}[Z] \geq -\mathbb{E}[Z]] \leq 2 \exp\left(\frac{-p_h^2 \eta^2 \binom{L}{2}^2 (C(D))^2}{2p_h C(D) \binom{U}{2}^2 + \frac{2}{3} C(D) \binom{U}{2}^2 p_h \eta}\right),$$

which is $\leq 1/n^c$ for $p_h = \widetilde{\Omega}\left(\frac{1}{\eta^2 C(D)}\right)$, when $U = \Theta(L)$. An equivalent proof applies for the lower tail $Y(D) \leq WDCC(D) - \eta$, after which we apply a union bound over the two tail bounds.

Now, we bound term (ii) of Eq. 5, which controls the error of our triangle-estimator. The proof is equivalent to the one for Theorem 3.6 by defining variables accordingly. Thus:

$$\left|\widehat{WDCC}(D) - Y(D)\right| = \frac{1}{\sum_{u, d_u \in D} \binom{d_u}{2}} \left|\widehat{A}(D) - A(D)\right| \leq \varepsilon Y(D),$$

with high probability. Combining item (i), (ii) we proved, for $D = [L, U]$ with $L \leq U \leq \tau$, that:

$$\begin{aligned} \left|\widehat{WDCC}(D) - WDCC(D)\right| &\leq \eta + \varepsilon Y(D) \leq \eta + \varepsilon(WDCC(D) + \eta) \\ &= \varepsilon \cdot WDCC(D) + (1 + \varepsilon)\eta. \end{aligned}$$

The final result from the theorem follows then by rescaling η .

□

Next, we prove bounds for our tail estimator, which provides estimates when, for $D = [L, U]$, we have $\tau \leq L$. Combining Theorem 3.8 and 3.9 yields Theorem 4.2 of main paper.

THEOREM 3.8. *Consider a degree interval $D = [L, U]$ such that $\tau \leq L < U$. For any $\varepsilon \in (0, 1/2)$, define $D^+ = [L(1-\varepsilon), U(1+\varepsilon)]$, and $D^- = [L(1+\varepsilon), U(1-\varepsilon)]$. For the quantity $\text{NDCC}(D) = \frac{S_{LCC}(D)}{C(D)}$, we have, if*

$$\tau = \tilde{\Omega}\left(\frac{1}{\varepsilon p_t}\right) \text{ and } p_{M_t} p_{A_t} = \tilde{\Omega}\left(\frac{\Delta_E(D^+)}{\varepsilon^2 T(D^-)}\right),$$

then Algorithm Estimate outputs a value $\widehat{\text{NDCC}}(D)$ such that:

$$\widehat{\text{NDCC}}(D) \in \left[(1-\varepsilon) \frac{S_{LCC}(D^-)}{C(D^+)}, (1+\varepsilon) \frac{S_{LCC}(D^+)}{C(D^-)} \right],$$

PROOF. For the tail estimator, it holds (w.h.p.) $\widehat{d}_v \in (1 \pm \varepsilon)d_v$ by Lemma 3.1 item (iii), for $\tau = \tilde{\Omega}\left(\frac{1}{\varepsilon p_t}\right)$. For the tail estimator, Alg. Estimate computes:

$$\widehat{\text{NDCC}}(D) = \frac{\sum_{v \in S_t, \lceil \widehat{d}_v \rceil \in D} \frac{\widehat{t}_v}{\binom{\widehat{d}_v}{2}}}{\left| \{v \in S_t, \lceil \widehat{d}_v \rceil \in D\} \right|} \stackrel{(i)}{=} \frac{\sum_{v, \lceil \widehat{d}_v \rceil \in D} \frac{\widehat{t}_v}{\binom{\widehat{d}_v}{2}}}{\left| \{v, \lceil \widehat{d}_v \rceil \in D\} \right|}, \quad (6)$$

where (i) follows from Lemma 3.1, item (iv) w.h.p. Define:

$$Y(D) = \frac{\sum_{v, \lceil \widehat{d}_v \rceil \in D} \frac{t_v}{\binom{\widehat{d}_v}{2}}}{\left| \{v, \lceil \widehat{d}_v \rceil \in D\} \right|},$$

and $\widehat{X}(D)$ using true degrees (in the binomial coefficients):

$$\widehat{X}(D) = \frac{\sum_{v, \lceil \widehat{d}_v \rceil \in D} \frac{t_v}{\binom{\widehat{d}_v}{2}}}{\left| \{v, \lceil \widehat{d}_v \rceil \in D\} \right|} = \frac{\widehat{S}_{LCC}(D)}{\widehat{C}(D)};$$

From Corollary 3.3, for $\tau \geq 4$, we have:

$$Y(D) \in \frac{1}{1 \mp 3\varepsilon} \widehat{X}(D). \quad (7)$$

Define $X(D)$ using true degree over sum and count as:

$$X(D) = \frac{\sum_{v, d_v \in D} \frac{t_v}{\binom{d_v}{2}}}{\left| \{v, d_v \in D\} \right|} = \frac{S_{LCC}(D)}{C(D)};$$

Now, we have the same terms in $\widehat{X}(D), X(D)$, but ranging on a different domain due to slacked (approximated) degrees. To explicit the domain differences, we consider complementary cdf to leverage monotonicity; that is, define $\widehat{F}(d) = \sum_{v, \lceil \widehat{d}_v \rceil \geq d} t_v / \binom{\widehat{d}_v}{2}$ and $F(d) = \sum_{v, d_v \geq d} t_v / \binom{d_v}{2}$. We have $\widehat{S}_{LCC}([L, U]) = \widehat{F}(L) - \widehat{F}(U)$, and $S_{LCC}([L, U]) = F(L) - F(U)$. By Lemma 3.1, items (ii) and (iii), and by rescaling ε 's, we have (w.h.p.):

$$\widehat{F}(d) \leq \sum_{v, d_v \geq \frac{d}{1+\varepsilon}} \frac{t_v}{\binom{d_v}{2}} \leq \sum_{v, d_v \geq (1-\varepsilon)d} \frac{t_v}{\binom{d_v}{2}} = F((1-\varepsilon)d),$$

and, similarly:

$$\widehat{F}(d) \geq \sum_{v, d_v \geq \frac{d}{1-\varepsilon}} \frac{t_v}{\binom{d_v}{2}} \geq \sum_{v, d_v \geq (1+\varepsilon)d} \frac{t_v}{\binom{d_v}{2}} = F((1+\varepsilon)d).$$

Consequently, for our estimator:

$$\begin{aligned} \widehat{S}_{LCC}([L, U]) &= \widehat{F}(L) - \widehat{F}(U) \leq F((1-\varepsilon)L) - F((1+\varepsilon)U) \\ &= S_{LCC}([(1-\varepsilon)L, (1+\varepsilon)U]), \end{aligned}$$

and, similarly:

$$\begin{aligned} \widehat{S}_{LCC}([L, U]) &= \widehat{F}(L) - \widehat{F}(U) \geq F((1+\varepsilon)L) - F((1-\varepsilon)U) \\ &= S_{LCC}([(1+\varepsilon)L, (1-\varepsilon)U]). \end{aligned}$$

By defining D^+, D^- as in the statement of the theorem, we have $\widehat{S}_{LCC}(D) \in [S_{LCC}(D^-), S_{LCC}(D^+)]$. With the same exact steps, we can derive $\widehat{C}(D) \in [C(D^-), C(D^+)]$. Combining with Eq. 7, we obtain:

$$Y(D) \in \frac{1}{1 \mp 3\varepsilon} \left[\frac{S_{LCC}(D^-)}{C(D^+)}, \frac{S_{LCC}(D^+)}{C(D^-)} \right].$$

Now, we bound $\widehat{\text{NDCC}}(D)$ with respect to $Y(D)$, which corresponds to the triangle-estimator error. This proof is equivalent to the one in Theorem 3.6. We just need to define w_Δ 's accordingly and notice that $w_\Delta \in \left[\frac{1}{\binom{U(1+\varepsilon)}{2}}, \frac{3}{\binom{L(1-\varepsilon)}{2}} \right]$. Applying the same steps leads to $\widehat{\text{NDCC}}(D) \in (1 \pm \varepsilon) Y(D)$ w.h.p. by setting $p_{M_t} p_{A_t} = \tilde{\Omega}\left(\frac{\Delta_E(D^+)}{\varepsilon^2 T(D^-)}\right)$. Notice that $T_t(D^-) = T(D^-)$ with high probability by Lemma 3.1 item (iv). Overall, we can write:

$$\widehat{\text{NDCC}}(D) \in \left[(1-\varepsilon) \frac{S_{LCC}(D^-)}{C(D^+)}, (1+\varepsilon) \frac{S_{LCC}(D^+)}{C(D^-)} \right],$$

as $\varepsilon \in (0, 1/2)$, and by rescaling, concluding the proof. $\square$

THEOREM 3.9. *Consider a degree interval $D = [L, U]$ such that $\tau \leq L < U$. For any $\varepsilon \in (0, 1/2)$, define $D^+ = [L(1-\varepsilon), U(1+\varepsilon)]$, and $D^- = [L(1+\varepsilon), U(1-\varepsilon)]$. For the quantity $\text{WDCC}(D) = \frac{T(D)}{W(D)}$, we have, if*

$$\tau = \tilde{\Omega}\left(\frac{1}{\varepsilon p_t}\right) \text{ and } p_{M_t} p_{A_t} = \tilde{\Omega}\left(\frac{\Delta_E(D^+)}{\varepsilon^2 T(D^-)}\right),$$

then Algorithm Estimate outputs a value $\widehat{\text{WDCC}}(D)$ such that:

$$\widehat{\text{WDCC}}(D) \in \left[(1-\varepsilon) \frac{T(D^-)}{W(D^+)}, (1+\varepsilon) \frac{T(D^+)}{W(D^-)} \right],$$

PROOF. The proof follows equivalent steps of the proof of Thm. 3.8, by applying complementary cdf on $T(D)$ and on $W(D)$. $\square$

Now, we prove our result on local clustering coefficients for sufficiently high-degree nodes (i.e., from our tail estimator).

THEOREM 4.3 FROM MAIN PAPER. *Let $v \in S_t$ be a node such that $d_v \geq \tau$ and $LCC(v) > 0$. For any $\varepsilon \in (0, 1/2)$ and any $\eta \in (0, LCC(v))$, if*

$$\tau = \tilde{\Omega}\left(\frac{1}{\varepsilon p_t}\right), \text{ and } p_{M_t} p_{A_t} = \tilde{\Omega}\left(\frac{\Delta_E(v) LCC(v)}{\eta^2 \tau^2}\right),$$

then, with high probability,

$$\left| \widehat{LCC}(v) - LCC(v) \right| \leq \varepsilon LCC(v) + \eta.$$

PROOF. First, a node v with degree $d_v \geq \tau$ is included in S_t for the above choice of τ with high probability, from Lemma 3.1. Our tail estimator computes, for any such node:

$$\widehat{LCC}(v) = \frac{\widehat{t}_v}{\binom{\lceil d_v \rceil}{2}} \stackrel{(i)}{=} \frac{1}{1 \mp \varepsilon} \frac{\widehat{t}_v}{\binom{d_v}{2}} := \frac{1}{1 \mp \varepsilon} Y(v),$$

where (i) follows from Corollary 3.3, and by rescaling ε . For vertex v , let $X_i(v) = 1/p_{\Delta_i}$ if i -th triangle is incident to v and is counted by the algorithm. We have $\mathbb{E}[X_i(v)] = 1$, so that $\widehat{t}_v = \sum_i X_i(v)$ and $\mathbb{E}[\widehat{t}_v] = t_v$. Moreover, $\text{Var}[X_i(v)] \leq 1/p_{\Delta_i} \leq 1/p_{APM}$, assuming $p_A \leq p_M$. We can build a dependency graph for variables $X_i(v)$ for all i , with fractional chromatic number $\leq \Delta_E(v)$, which is the maximum number of triangles (among those incident to v) adjacent to an edge. Define centered variables $Z_i(v) = X_i(v) - \mathbb{E}[X_i(v)]$, and $Z = \sum_i Z_i(v)$. By Lemma 1.5:

$$\begin{aligned} \mathbb{P}[|Y(v) - LCC(v)| \geq \eta] &= \mathbb{P}\left[|Z| \geq \eta \binom{d_v}{2}\right] \\ &\leq 2 \exp\left(\frac{-8\eta^2 \binom{d_v}{2}^2}{25\Delta_E(v) \left(\frac{t_v}{p_{APM}} + \frac{\eta \binom{d_v}{2}}{3p_{APM}}\right)}\right), \end{aligned}$$

which is $\leq 1/n^c$ for $p_{APM} = \widetilde{\Omega}\left(\frac{\Delta_E(v)LCC(v)}{\eta^2 \tau^2}\right)$ for $\eta \leq LCC(v)$, and since $d_v \geq \tau$. Overall:

$$\begin{aligned} |\widehat{LCC}(v) - LCC(v)| &\leq |\widehat{LCC}(v) - Y(v)| + |Y(v) - LCC(v)| \\ &\leq \varepsilon Y(v) + \eta \leq \varepsilon LCC(v) + \eta(\varepsilon + 1). \end{aligned}$$

Main result follows by rescaling ε and η . $\square$

Now, we report the proof of the equivalence of the tail triangle estimator in the two-pass and in the one-pass algorithms.

LEMMA 5.1 FROM MAIN PAPER. *The estimators produced by the one-pass and two-pass algorithms are identical for the node-degree map S_t and, if $p_{M_t} = p_{A_t}$, also for the node-triangle map $\widehat{T}_t$.*

PROOF. We prove the lemma by coupling the randomness of the one-pass and two-pass algorithms. For each edge $e = \{u, v\}$ in the stream, independently pre-generate Bernoulli random variables X_u, X_v with parameter p_t , used to determine whether $u, v \in S_t$, and a Bernoulli random variable Y_e with parameter $p_{M_t} = p_{A_t}$, used to determine whether e is sampled for tail triangle counting. Consider the same variables $\{X_u, X_v, Y_e\}_{e=\{u,v\}}$ for both algorithms.

Since both algorithms use the same Bernoulli variables X_u, X_v to decide whether nodes u and v belong to S_t , the resulting set S_t is identical in the two algorithms.

Now, fix any node $v \in S_t$, and consider any triangle of the type $\Delta = \{\{v, a\}, \{v, b\}, \{a, b\}\}$ incident to v . Let e_1, e_2 be the two edges of Δ that appear before the closing edge in the stream.

We first analyze the two-pass algorithm. After the first pass, the set S_t is fixed. An edge e is inserted into the tail main sample if it is incident to a node of S_t and $Y_e = 1$; otherwise, if $Y_e = 1$, it is inserted into the tail auxiliary sample. Since $v \in S_t$, at least one of the two edges e_1, e_2 is incident to v . Hence, whenever that edge is sampled, it must belong to the tail main sample. Therefore, the

event that Δ is counted by the two-pass algorithm is exactly $Y_{e_1} = 1$ and $Y_{e_2} = 1$.

Consider now the one-pass algorithm. At the arrival time of an edge e , the algorithm decides, based on the current stream prefix, whether e is eligible for the tail main sample or for the tail auxiliary sample. Since $p_{M_t} = p_{A_t}$, in both cases the decision is made using the same variable Y_e . Thus, independently of whether e is currently being sampled in the main or auxiliary sample, the event that e is sampled is $Y_e = 1$. The one-pass algorithm counts a triangle whenever its two non-closing edges are sampled, even if both are stored in the auxiliary sample. Hence, for the same triangle Δ , the event that it is counted is again exactly $Y_{e_1} = 1$ and $Y_{e_2} = 1$.

Thus, for every triangle incident to $v \in S_t$, the counting event is identical in the one-pass and two-pass algorithms. Summing over all such triangles shows that the number of counted triangles incident to v is the same in the two algorithms, for every $v \in S_t$. $\square$

Finally, we present the proofs about space and time complexities of our proposed one-pass algorithm.

THEOREM 5.2 FROM MAIN PAPER. *Let $\mathcal{D} = \{D_1, D_2, \dots\}$ be a set of degree intervals. There exists a one-pass streaming algorithm that, for all $D = [L, U) \in \mathcal{D}$ simultaneously, computes approximations of $\text{NDCC}(D)$ and of $\text{WDCC}(D)$ using expected space:*

$$\begin{aligned} \widetilde{O}\left(\frac{n}{\eta^2} \max_{D: U \leq \tau} \left(\frac{1}{C(D)} + \frac{\Delta_V(D)}{T(D)}\right) + \frac{m}{\varepsilon \tau}\right) \text{ nodes, and} \\ \widetilde{O}\left(\frac{m}{\varepsilon} \left(\eta + \max_D \sqrt{\frac{\Delta_E(D^+)}{T(D^-)}}\right)\right) \text{ edges,} \end{aligned}$$

where $\tau = \frac{1}{\varepsilon p_t}$.

PROOF. The overall pipeline is as follows: first, run our one-pass algorithm to gather node-degree and node-triangle estimates. Parameters of the one-pass algorithm should be fixed accordingly to $\mathcal{D}$: if all $D = [L, U) \in \mathcal{D}$ are such that $\tau \leq L$, then all intervals are estimated by using the tail estimator (i.e., there is no point in using the head estimator and hence one can set $p_h = 0$). In general, p_h should consider only degrees $D = [L, U) \in \mathcal{D}$: $U \leq \tau$. Note that $\tau = 1/(\varepsilon p_t)$ can be computed upfront, given our fixed parameter p_t . Upon running the one-pass algorithm, for every degree interval $D = [L, U) \in \mathcal{D}$ run Algorithm Estimate to obtain estimates $\widehat{\text{NDCC}}(D)$ and $\widehat{\text{WDCC}}(D)$, either using the head estimator (if $U \leq \tau$) or the tail estimator (if $\tau \geq L$).

Consider a fixed interval $D \in \mathcal{D}$. We plug bounds derived in Thm. 3.6, 3.7, 3.8, 3.9. The space used by the algorithm accounts for the node-degree map S_h (head) and S_t (tail), and the subgraph stored for counting triangles. The expected number of nodes in S_h is $n \cdot p_h$, and the expected number of nodes in S_t is $\sum_v (1 - (1 - p_t)^{d_v}) \leq \sum_v \min(1, p_t d_v)$. We consider the looser $mp_t = m/(\varepsilon \tau)$ for the sake of interpretability of our bound. The expected number of edges for counting triangles is bounded by $m(p_{M_h} + p_{A_h} + p_{M_t} + p_{A_t})$, subject to constraints $p_{M_h} p_{A_h} = \widetilde{O}\left(\frac{1}{\varepsilon^2} \left(\eta^2 + \frac{\Delta_E(D)}{T(D)}\right)\right)$, and $p_{M_t} p_{A_t} = \widetilde{O}\left(\frac{\Delta_E(D^+)}{\varepsilon^2 T(D^-)}\right)$. Thus, the quantity $p_{M_h} + p_{A_h} + p_{M_t} + p_{A_t}$ is minimised when $p_{M_h} = p_{A_h} = \widetilde{O}\left(\sqrt{\frac{1}{\varepsilon^2} \left(\eta^2 + \frac{\Delta_E(D)}{T(D)}\right)}\right)$ and $p_{M_t} =$

$p_{A_t} = \tilde{O}\left(\sqrt{\frac{\Delta_F(D^+)}{\varepsilon^2 T(D^-)}}\right)$. We have $p_h = \tilde{\Omega}\left(\frac{1}{\eta^2}\left(\frac{1}{C(D)} + \frac{\Delta_V(D)}{T(D)}\right)\right)$, and $\frac{T(D)}{C(D)} \leq \Delta_V(D)$. Plugging and re-arranging, yields the final bound for a single degree interval $D \in \mathcal{D}$. The result follows by a union bound on $|\mathcal{D}| = O(n^2)$ and considering the interval D which maximizes each quantity depending on it. Again, by prioritizing clarity of our bounds, we do not make separation between head and tail intervals in our edge space. $\square$

PROPOSITION 1 FROM MAIN PAPER. *Given an input stream Σ for graph G with m edges and maximum degree $d_{\max}(G)$, our one-pass algorithm processes the stream in $O(m \cdot d_{\max}(G))$ time.*

PROOF. For each incoming edge $\{u, v\}$ in the stream Σ , the most expensive step is the computation of common neighbors when counting triangles. For edge $\{u, v\}$, computing common neighbors takes $O(\min(d_u, d_v))$, where adjacency lists of subgraph are implemented by hash sets, so we can iterate over the smaller set and use set membership in the largest one. Iterating over all the edges, it trivially follows $\sum_{\{u,v\} \in \Sigma} O(\min(d_u, d_v)) = O(m \cdot d_{\max}(G))$, where $d_{\max}(G)$ is the maximum degree for graph G , and is a rough upper bound as we need to consider degrees in the subgraph induced by sampled edges. All remaining operations, including hash computations, accesses to the node-degree maps S_h, S_t , and insertions into M_h, M_t, A_h , and A_t , take constant time per operation under standard hashing assumptions. The final step for correcting tail nodes takes at most $O(m)$ time, and is implemented by guessing the value r through binary search. $\square$

4 SELECTING PARAMETERS FOR BOLIDE

In the following, we provide analytical insights into the parameter selection strategy used in our experiments. Our algorithm BOLIDE allocates memory for counting triangles across four different reservoirs: B_{M_h} for head main sample, B_{A_h} for head auxiliary sample, B_{M_t} for tail main sample, and B_{A_t} for tail auxiliary sample.

Head Main and Auxiliary Sample. Given the probability p_h of sampling head nodes, which determines the sample S_h , let $m_{M_h} = |\{\{u, v\} \in E : u \in S_h \text{ or } v \in S_h\}|$ (number of candidate edges for head main sample), and $m_{A_h} = |\{\{u, v\} \in E : u \notin S_h \text{ and } v \notin S_h\}|$ (number of candidate edges for head auxiliary sample). In expectation, we have $m_{M_h} = (2p_h - p_h^2)m$ and $m_{A_h} = (1 - p_h)^2 m$.

Consider the counting of a triangle $\{u, v, w\}$ with at least one vertex (say v) belongs to S_h . We can have:

- $u, w \notin S_h$: depending on the arrival order, the two edges (when the closing one arrives) can be both in the main sample or one in the main and the other in the auxiliary sample. Assuming random stream ordering, the former happens w.p. $1/3$, and the latter w.p. $2/3$.
- $u \in S_h$ or $w \in S_h$: no matter the arrival order, both edges (when the closing one arrives) are in the main sample.

Thus, for the given triangle, the probability that both edges are in the main sample is $\frac{1}{3}(1 - p_h)^2 + \left(1 - (1 - p_h)^2\right) = 1 - \frac{2}{3}(1 - p_h)^2$, while the probability that one edge is in the main and the other in the auxiliary sample is $\frac{2}{3}(1 - p_h)^2$.

Following previous work [6, 11], we consider the simplified variance (i.e., ignoring covariance), as it has been shown that it is strongly correlated with the variance in real-world graphs ($R^2 > 0.99$). Define the “effective” sampling probabilities $q_M = B_{M_h}/m_{M_h}$ and $q_A = B_{A_h}/m_{A_h}$ from our reservoir samples. Minimizing the simplified variance implies minimizing the following upper bound on each individual triangle variance¹:

$$V = \frac{1}{q_M^2} \left(1 - \frac{2}{3}(1 - p_h)^2\right) + \frac{1}{q_M q_A} \left(\frac{2}{3}(1 - p_h)^2\right). \quad (8)$$

Our objective is to minimize V subject to a fixed memory budget for head samples $k_h = B_{M_h} + B_{A_h}$. We know that, in expectation, $B_{M_h} = q_M \cdot m_{M_h}$ and $B_{A_h} = q_A \cdot m_{A_h}$, for fixed probability sampling. We plug in the expected quantity² in (8)

Let $y = B_{M_h}/k_h$ be the budget allocation for head main sample. Setting $\frac{dV}{dy} = 0$ we obtain a quadratic equation $ay^2 + by + c = 0$ with: $a = -\frac{4}{3} + \frac{20}{3}p_h - \frac{10}{3}p_h^2$, $b = \frac{2}{3} - \frac{16}{3}p_h - \frac{16}{3}p_h^2 + 8p_h^3 - 2p_h^4$, $c = \frac{4}{3}p_h + \frac{14}{3}p_h^2 - \frac{16}{3}p_h^3 + \frac{4}{3}p_h^4$. Consequently, the optimal parameter selection can be computing deterministically given p_h by solving the above quadratic equation. For instance, when $p_h = 0.1$, this selection yields $B_{M_h} \approx 0.56k_h$ and $B_{A_h} \approx 0.44k_h$; for $p_h = 0.2$, it shifts to $B_{M_h} \approx 0.65k_h$ and $B_{A_h} \approx 0.35k_h$.

Tail Main and Auxiliary Sample. Given that tail node sampling is degree-dependent, providing an analytical closed-form budget allocation (as done for the head component) is intractable. Remember that tail triangles are counted even when their edges arrive prior to the sampling of tail nodes, in which case both edges belong to the tail auxiliary sample (see Section 5 of main paper).

To resolve the tail budget allocation, we assume that a triangle is equally likely to be captured in one of the three following configurations: (i) entirely within the main sample, (ii) entirely within the auxiliary sample, or (iii) split across both. By similar calculations, an upper bound on the simplified variance is $\frac{1}{3q_M^2} + \frac{1}{3q_M q_A} + \frac{1}{3q_A^2}$ for $q_M = B_{M_t}/k_t$ and $q_A = B_{A_t}/k_t$ and subject to $k_t = B_{M_t} + B_{A_t}$, which is minimized setting $B_{M_t} = B_{A_t}$, i.e., equally balancing the tail main and auxiliary budgets.

5 ADDITIONAL EXPERIMENTS

In this section, we first assess empirically the quality of bounds obtained by our tail estimator. Moreover, we analyze the extra number of counters necessary for guaranteeing correct estimates for our final one-pass algorithm. Then, we expand on the empirical choice of parameters for our algorithm BOLIDE, and finally we report results missing in the main paper due to space issues.

5.1 Empirical Bounds for Tail Estimator

In the following, we empirically assess the quality of the bounds reported by our tail estimator. For degree interval D , let $\text{NDCC}(D) = S_{LCC}(D)/C(D)$. From Theorem 3.8, Algorithm Estimate outputs

¹Technically, the probability is $B_{M_h}(B_{M_h} - 1)/(m_{M_h}(m_{M_h} - 1))$ rather than q_M^2 ; we assume equivalence for sufficiently large numbers.

²The derivation is not formal for the ease of presentation. Standard concentration inequalities (e.g., Hoeffding’s, see 1.2) guarantee that the empirical count matches expectations with high probability.

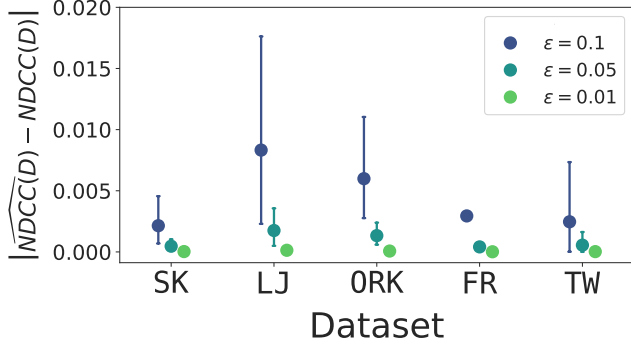


Figure 1: Absolute error of estimates given by bound in Eq. 9, for various values of ε . For each dataset and ε , the average and standard deviation over 20 different intervals $D = [2^i, 2^{i+1})$, such that $\tau \leq 2^i$, are shown.

estimate $\widehat{NDCC}(D)$ such that:

$$\widehat{NDCC}(D) \in \left[(1 - \varepsilon) \frac{S_{LCC}(D^-)}{C(D^+)}, (1 + \varepsilon) \frac{S_{LCC}(D^+)}{C(D^-)} \right], \quad (9)$$

for any degree interval $D = [L, U)$ such that $L \geq \tau$, where $D^- = [(1 + \varepsilon)L, (1 - \varepsilon)U]$ and $D^+ = [(1 - \varepsilon)L, (1 + \varepsilon)U]$.

To empirically quantify such bounds, we compute absolute error $|\widehat{NDCC}(D) - NDCC(D)|$ considering degree intervals of powers of two (i.e., $D = [2^i, 2^{i+1})$). We consider values of $\widehat{NDCC}(D)$ as the middle point with respect to the interval guarantees from Equation 9. We pick 20 intervals equally spaced, making sure that for each interval $D = [L, U)$ we have $L \geq \tau$ (i.e., when we employ our tail estimator), for values of τ computed by our algorithm BOLIDE in the experiments (see main paper for some representative values of degree threshold τ).

Figure 1 shows results for $\varepsilon \in \{0.01, 0.05, 0.1\}$. While for $\varepsilon = 0.1$ we observe large absolute errors and high variance, lower values of ε (i.e., tighter intervals D^-, D^+) allow for very accurate and precise stable bounds, across all the datasets and the considered intervals. This means that in real-world networks, the distribution of degree-binned clustering coefficient is stable when allowing small slack in the binned intervals (for high degrees), and our bounds in Theorem 3.8 are empirically accurate.

5.2 One Pass Algorithm: Extra Counters

In this section, we investigate the number of extra counters needed to ensure unbiasedness of estimates for our one-pass algorithm. Let N be the number of nodes for which we know the degree (i.e., $|S_t|$), and N' be the number of nodes for which we know the number of local triangles (i.e., $|\hat{T}_t|$). In our two-pass algorithm, we have $N \geq N'$, given that our second pass SecondPass is estimating triangles only incident to nodes for which we know the degree from the first pass FirstPass. Conversely, our one-pass algorithm also counts triangles incident to nodes for which (currently) we are unaware of their degrees. Such nodes belong to edges in the auxiliary sample; counting such triangles allows for the unbiasedness (see Section 5 of main paper). That is, for our one-pass algorithm, we may have $N \leq N'$. Note that the extra counters for triangles will be wasted,

Table 1: Number $c = W/N$ of extra counters stored by our one-pass algorithm, where N, N' are the number of nodes for which we estimate their degree, and the number of nodes for which we estimate triangles, and $W = \max(0, N' - N)$ is the number of “wasted” counters. For each dataset and choice of auxiliary sample budget, the average and standard deviation over 10 independent repetitions are reported.

Dataset	Tail Auxiliary Sample Budget B_{A_t}		
	0.005m	0.01m	0.015m
SK	0.00 $\pm$ 0.00	0.22 $\pm$ 0.04	0.55 $\pm$ 0.03
LJ	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.28 $\pm$ 0.00
ORK	0.81 $\pm$ 0.00	1.58 $\pm$ 0.01	2.24 $\pm$ 0.01
TW	0.40 $\pm$ 0.01	0.73 $\pm$ 0.01	1.03 $\pm$ 0.01
FR	0.11 $\pm$ 0.00	0.53 $\pm$ 0.00	0.92 $\pm$ 0.00

as we cannot provide an estimate of clustering coefficients without knowing degrees (or estimates of degrees) for such nodes.

Table 1 shows the fraction of extra counters stored by our one-pass algorithm. We compute the number W of “wasted” counters as $W = \max(0, N' - N)$ for each dataset, and for different size of tail auxiliary sample $B_{A_t} = f \cdot m$ for values of $f \in \{0.005, 0.01, 0.015\}$. The other parameters are as explained in Section 5.4. We observe that, across all datasets, the wasted counters range from 0 to 2.24 times the number of counters effectively needed.

5.3 Empirical Values: $\Delta_V(D)$, $\Delta_E(D)$, $T(D)$

In the following, we inspect values of $\Delta_V(D)$, $\Delta_E(D)$ and $T(D)$ in real-world datasets. Recall that, for a degree interval D , $\Delta_V(D)$ is the maximum number of triangles incident to a node having degree in D , and $\Delta_E(D)$ is the maximum number of triangles (among those incident to nodes in V_D) adjacent to an edge.

Table 2 shows values which appear in our space bound, considering intervals of power of two (i.e., $D = [2^i, 2^{i+1})$) on SK, LJ and ORK. We compute slacked intervals $D^- = [L(1 + \varepsilon), U(1 - \varepsilon)]$ and $D^+ = [L(1 - \varepsilon), U(1 + \varepsilon)]$ for any $D = [L, U)$ reported in the first column of the table, considering $\varepsilon = 0.05$. We observe that the ratio $\Delta_E(D^+)/T(D^-)$, which our edge-space depends on, is consistently very small across all datasets: its maximum is below 0.009 for all the considered intervals and datasets.

5.4 Choice of Parameters

We explain the choice of parameters for our algorithm BOLIDE. Overall, BOLIDE depends on several parameters: probabilities p_h and p_t of sampling head and tail nodes, memory budgets B_{M_h}, B_{M_t} for head and tail main samples, and memory budgets B_{A_h}, B_{A_t} for the auxiliary samples. Clearly, the higher each of these parameters is, the better the estimates, as they all govern the space used.

We perform a grid search to identify the best choice of parameters. We consider values of $p_h \in \{0.05, 0.1, 0.15, 0.2, 0.25, 0.3\}$ and $p_t \in \{0.025, 0.05, 0.075\} \cdot n/m$, and we constrain BOLIDE to use exactly 10% m edges, considering head-tail allocation $(B_{M_h} + B_{A_h}, B_{M_t} + B_{A_t}) = \{(0.01, 0.09), (0.05, 0.05), (0.09, 0.01)\} \cdot m$. Consequently, we set main-auxiliary allocation as explained in Sect. 4:

Table 2: Table reporting, for each degree interval $D = [2^i, 2^{i+1})$, the quantities $C(D)$ (number of nodes with degree in D), $T(D)$ (number of triangles incident to nodes with degree in D), $\Delta_V(D)$ (maximum number of triangles incident to a node with degree in D), $T(D^-)$ and $\Delta_E(D^+)$ (maximum number of triangles incident to an edge containing a node with degree in D^+). We set $\varepsilon = 0.05$ for computing intervals $D^- = [L(1 + \varepsilon), U(1 - \varepsilon))$ and $D^+ = [L(1 - \varepsilon), U(1 + \varepsilon))$.

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$[2^i, 2^{i+1})$	$C(D)$	$T(D)$	$T(D^-)$	$\Delta_V(D)$	$\Delta_E(D^+)$	$C(D)$	$T(D)$	$T(D^-)$	$\Delta_V(D)$	$\Delta_E(D^+)$	$C(D)$	$T(D)$	$T(D^-)$	$\Delta_V(D)$	$\Delta_E(D^+)$
$[2^1, 2^2)$	447k	291k	194k	3	1.4k	919k	756k	501k	3	218	81.6k	65.8k	46.2k	3	102
$[2^2, 2^3)$	456k	1.4M	1.2M	21	8.5k	834k	3.5M	2.9M	21	188	148k	595k	517.7k	21	117
$[2^3, 2^4)$	319k	4.0M	3.6M	105	9.8k	743k	11M	10.1M	105	223	300k	4.3M	4.0M	105	493
$[2^4, 2^5)$	150k	6.7M	5.9M	429	14k	617k	27.7M	25M	465	452	555k	25.0M	22.1M	433	1.7k
$[2^5, 2^6)$	69k	8.8M	7.7M	1.5k	18k	404k	51.5M	45.7M	2.0k	872	772k	100M	87.1M	1.6k	3.2k
$[2^6, 2^7)$	21.8k	7.0M	6.0M	4.3k	952	185k	70.9M	62.8M	7.9k	1.3k	685k	247M	221M	5.3k	3.0k
$[2^7, 2^8)$	9.4k	7.7M	6.7M	10.1k	524	62.3k	110M	97.8M	31.8k	1.2k	333k	326M	298M	16.0k	2.6k
$[2^8, 2^9)$	3.7k	8.4M	7.6M	18.3k	520	19.4k	158M	144M	114k	715	96.2k	239M	210M	30.2k	2.2k
$[2^9, 2^{10})$	1.4k	9.0M	7.8M	34.3k	1.0k	4.9k	91.8M	81.7M	214k	1.0k	29.7k	199M	186M	57.9k	2.4k
$[2^{10}, 2^{11})$	536	5.7M	4.7M	69.3k	2.1k	704	18.4M	15.1M	229k	1.4k	3.0k	75.4M	65.5M	118k	1.7k
$[2^{11}, 2^{12})$	211	4.5M	4.2M	107k	4.1k	161	4.6M	3.9M	123k	2.5k	735	48.8M	41.7M	266k	3.1k
$[2^{12}, 2^{13})$	85	2.7M	2.0M	154k	8.2k	29	1.8M	1.7M	123k	4.3k	250	34.9M	31.8M	591k	5.9k
$[2^{13}, 2^{14})$	60	3.6M	3.2M	283k	16.2k	11	1.3M	1.2M	368k	4.5k	99	28.1M	24.8M	1.0M	9.1k
$[2^{14}, 2^{15})$	24	3.5M	3.3M	531k	28.7k	1	59k	0	59k	840	27	16.8M	16.3M	1.7M	9.1k

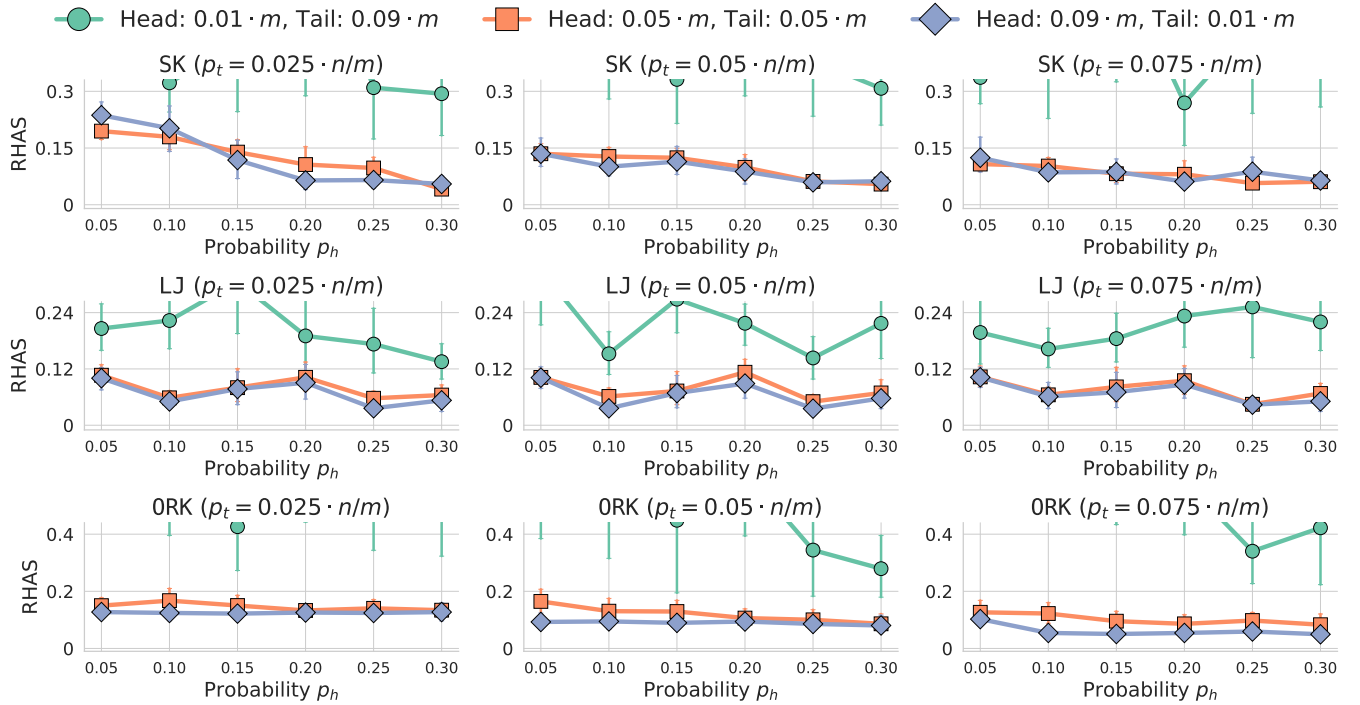


Figure 2: RHAS distance between exact and estimated NDCC distribution, for degree intervals of powers of two (i.e., $D = [2^i, 2^{i+1})$) versus probability p_h of sampling head nodes. For each row (dataset), each subplot shows a different value of probability p_t of sampling tail nodes. For each dataset and choice of parameters, the average and 95% confidence intervals over 10 independent repetitions are shown. y -axis is shared across rows (datasets).

solving the quadratic equation based on p_h for head budgets B_{M_h} and B_{A_h} , and splitting equally tail budgets $B_{M_t} = B_{A_t}$.

Given that these experiments do not involve scalability of our algorithm, we do not consider our biggest datasets (TW and FR).

Figure 2 depicts results; x -axis shows values of p_h , and y -axis shows RHAS for distribution $\{NDCC(D_i)\}_i$ and degree intervals of powers of two (i.e., $D = [2^i, 2^{i+1})$). For each row (dataset), a subplot represents a different value of p_t . We see that the choice

$B_{M_h} + B_{A_h} = 0.09m$, $B_{M_t} + B_{A_t} = 0.01m$ is consistently the best and the most stable, while the opposite assignment $B_{M_h} + B_{A_h} = 0.01m$, $B_{M_t} + B_{A_t} = 0.09m$ is performing badly and exhibits high variance. (It is cropped from the figure for the sake of visualization.) The reason is that, in real-world datasets, degree distribution is heavy-tailed: there are many more edges incident to low-degree nodes (as they are a lot) conversely to high-degree nodes (as they are few). Hence, giving significantly more budget to the head region is the right move.

The best and most robust choice of probabilities for sampling nodes would be $p_h = 0.3$ and $p_t = 0.075n/m$, i.e., the highest values. However, this implies to store node-degrees counters for $45\%n$, on average. We thus opted for the more conservative choice $p_h = 0.2$ and $p_t = 0.05n/m$, which provides similar accuracy while tracking degrees only for a 30% fraction of the nodes. In any case, we note that our algorithm BOLIDE is very robust for different choices of p_h and p_t , with the head-tail split of $0.09m$, $0.01m$.

5.5 Comparison with baselines

We compare BOLIDE with TRIEST-IMPR and WRS baselines. In the following, we provide a brief description of baselines and how we implemented their degree estimator.

- TRIEST-IMPR [9] works by sampling edges via (a unique) reservoir sampling of size equal to a given memory budget. Consider the sample $\mathcal{R}$ of edges in the reservoir at the end of the stream. We output the local triangle estimates $\hat{t}_u$ for each node $u \in \mathcal{R}$, together with the unbiased degree estimator $\hat{d}_u = \frac{m}{|\mathcal{R}|} |\{u, v\} \in \mathcal{R}\}|$. Estimates for NDCC are then computed by averaging LCCs for each degree interval. Note that reservoir sampling is an edge-sampling scheme, and thus keep nodes with probability proportional to their degree. As shown in the main paper, TRIEST-IMPR fails to provide estimate for low-degree nodes, as they are unrepresented in the sample.
- WRS [6, 7] couples reservoir sampling with waiting room, which stores the most recent edges in the stream. As suggested in their publication, we set the waiting room size to be a 0.1 fraction of the total budget. Consider the sample $\mathcal{R} \cup \mathcal{W}$ of edges in the reservoir and waiting room at the end of the stream. We output the local triangle estimates $\hat{t}_u$ for each node $u \in \mathcal{R} \cup \mathcal{W}$, together with the unbiased degree estimator $\hat{d}_u = \frac{m}{|\mathcal{R}|} |\{u, v\} \in \mathcal{R}\}| + |\{u, v\} \in \mathcal{W}\}|$. Because of the bias on low-degree nodes, which are counted only within the waiting room, we cap the average of local clustering coefficients to one, for each degree interval. Estimates for NDCC are then computed by averaging LCCs for each degree interval. As shown in the main paper, WRS fails to provide estimate for low-degree nodes, as they are both unrepresented in the reservoir sampling and they are not captured precisely within the waiting room, which is useful for counting global triangles rather than clustering coefficients.

5.6 Additional Results

We report all the results missing from Section 6 of the main paper. We display estimates of BOLIDE for $\{NDCC(D_i)\}_i$ and $\{WDCC(D_i)\}_i$ distributions, considering degree intervals $D_i = [b^i, b^{i+1})$, for $b = \{2, 1.5, 1.2, 1.1\}$. Figures 3 and 4 depict results for NDCC for $b =$

$\{2, 1.1\}$, while Figures 5 and 6 show results for WDCC for $b = \{2, 1.5, 1.2, 1.1\}$. Similarly to what was observed in Section 6 of the main paper, we see that BOLIDE is qualitatively good, as the estimated distributions of NDCC and WDCC are very close to the exact distribution. Moreover, our algorithm is quantitatively accurate given that RHAS distance is consistently well below 15%.

5.7 Comparison with ECC

In the following, we provide a more detailed comparison between algorithms ECC and BOLIDE when varying the number K of iterations used by ECC. Remember that ECC runs K parallel copies of a sparsified graph with sparsification rate equal to $1/C$, where C is the number of colors. As in the experiments in the main text, we fix memory budget to $0.1m$ for both algorithms, and we run algorithm ECC for number K of iterations in $[50, 100, 150, 200, 250, 300, 350, 400]$, as in [5]. The number C of colors is then $C = K/0.1$.

Clearly, the lower the iterations, the higher the sparsification rate. This implies that ECC retrieves more nodes (of lower degrees), and evaluation of LCCs is performed with respect to a bigger set H .

Figure 8 shows results, for our smallest datasets SK, LJ and ORK. We observe that BOLIDE is significantly better than ECC for each number $K > 50$ of iterations and for each metric on SK and LJ, and comparable for $K = 50$, while on ORK our algorithm is better for $K > 100$. For small K 's, ECC uses higher sparsification rate and retrieves nodes of lower degree, which may be harmful for BOLIDE as they are closer to the degree threshold τ . Remember that BOLIDE has theoretical guarantees for LCCs only in the tail estimator. Notably, BOLIDE employs only the tail estimator for giving estimates of LCCs, which consumes $0.01m$, while ECC consumes in expectation $0.1m$.

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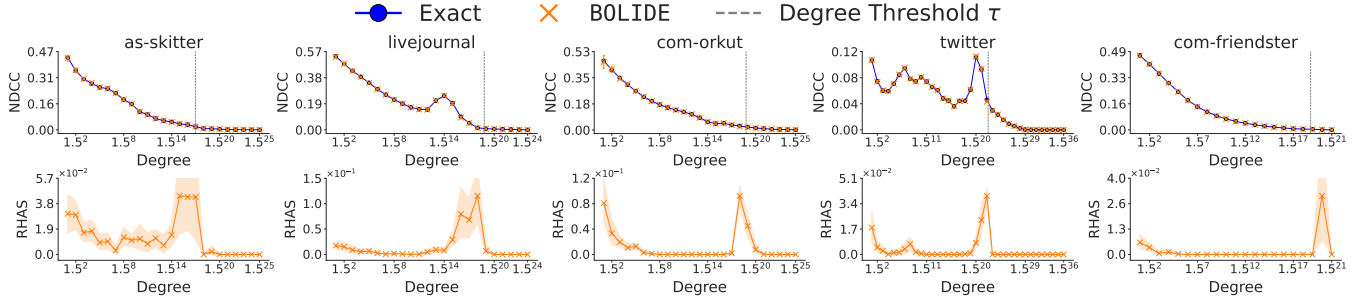


Figure 3: Exact and estimated distributions for NDCC (top) and pointwise RHAS distance between them (bottom), for degree intervals of powers of 1.5 (i.e., $D_i = [1.5^i, 1.5^{i+1})$). For each dataset, the average and standard deviation over 10 independent repetitions are shown. Dashed red line indicates the (average) degree threshold τ , separating head and tail estimators of BOLIDE.

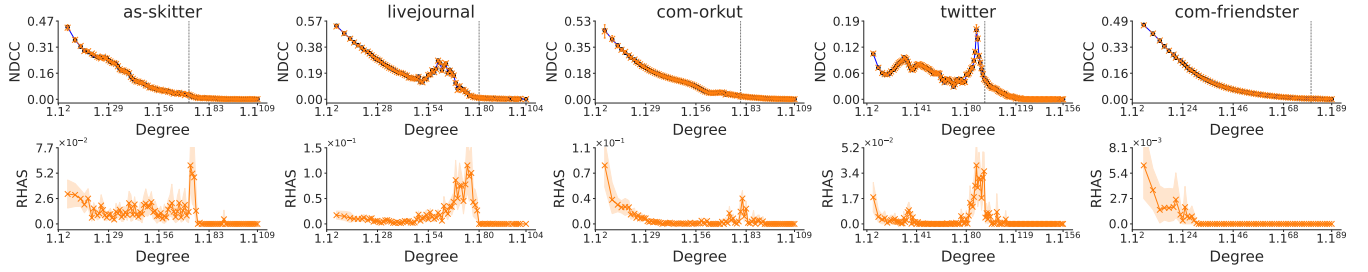


Figure 4: Exact and estimated distributions for NDCC (top) and pointwise RHAS distance between them (bottom), for degree intervals of powers of 1.1 (i.e., $D_i = [1.1^i, 1.1^{i+1})$). For each dataset, the average and standard deviation over 10 independent repetitions are shown. Dashed red line indicates the (average) degree threshold τ , separating head and tail estimators of BOLIDE.

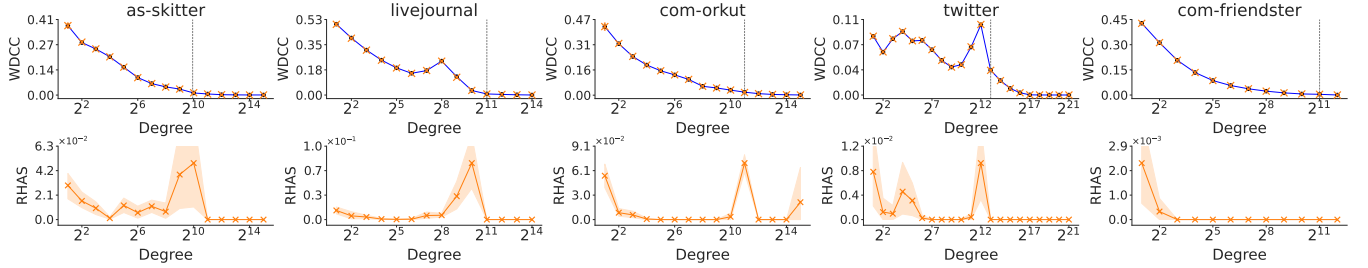


Figure 5: Exact and estimated distributions for WDCC (top) and pointwise RHAS distance between them (bottom), for degree intervals of powers of 2 (i.e., $D_i = [2^i, 2^{i+1})$). For each dataset, the average and standard deviation over 10 independent repetitions are shown. Dashed red line indicates the (average) degree threshold τ , separating head and tail estimators of BOLIDE.

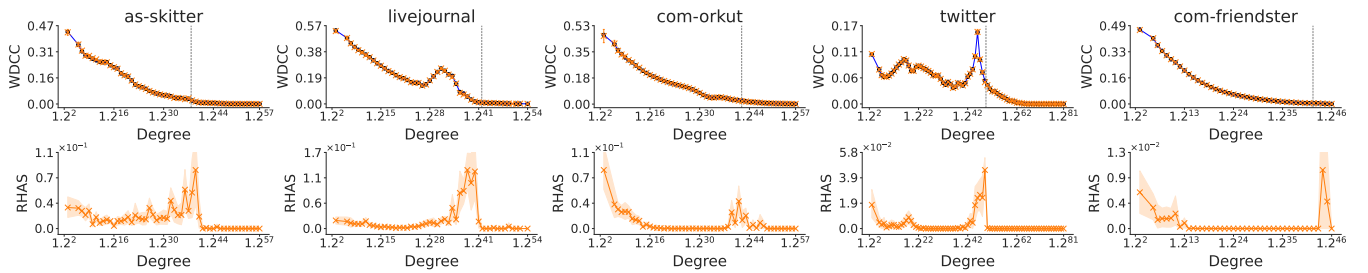


Figure 6: Exact and estimated distributions for WDCC (top) and pointwise RHAS distance between them (bottom), for degree intervals of powers of 1.2 (i.e., $D_i = [1.2^i, 1.2^{i+1})$). For each dataset, the average and standard deviation over 10 independent repetitions are shown. Dashed red line indicates the (average) degree threshold τ , separating head and tail estimators of BOLIDE.

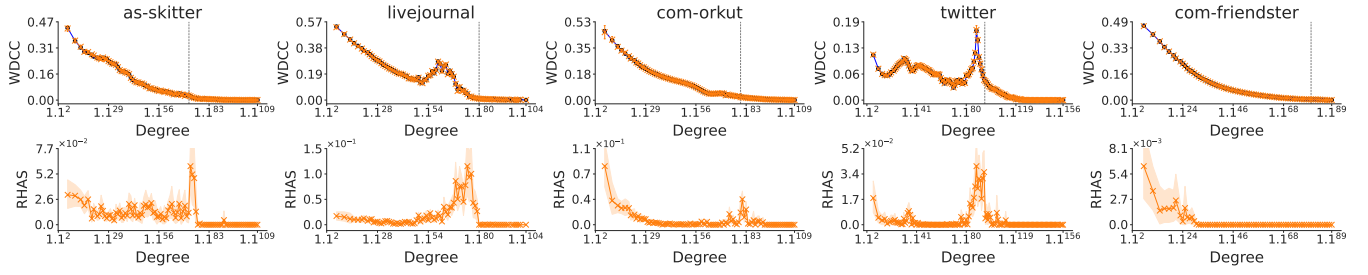


Figure 7: Exact and estimated distributions for WDCC (top) and pointwise RHAS distance between them (bottom), for degree intervals of powers of 1.1 (i.e., $D_i = [1.1^i, 1.1^{i+1})$). For each dataset, the average and standard deviation over 10 independent repetitions are shown. Dashed red line indicates the (average) degree threshold τ , separating head and tail estimators of BOLIDE.

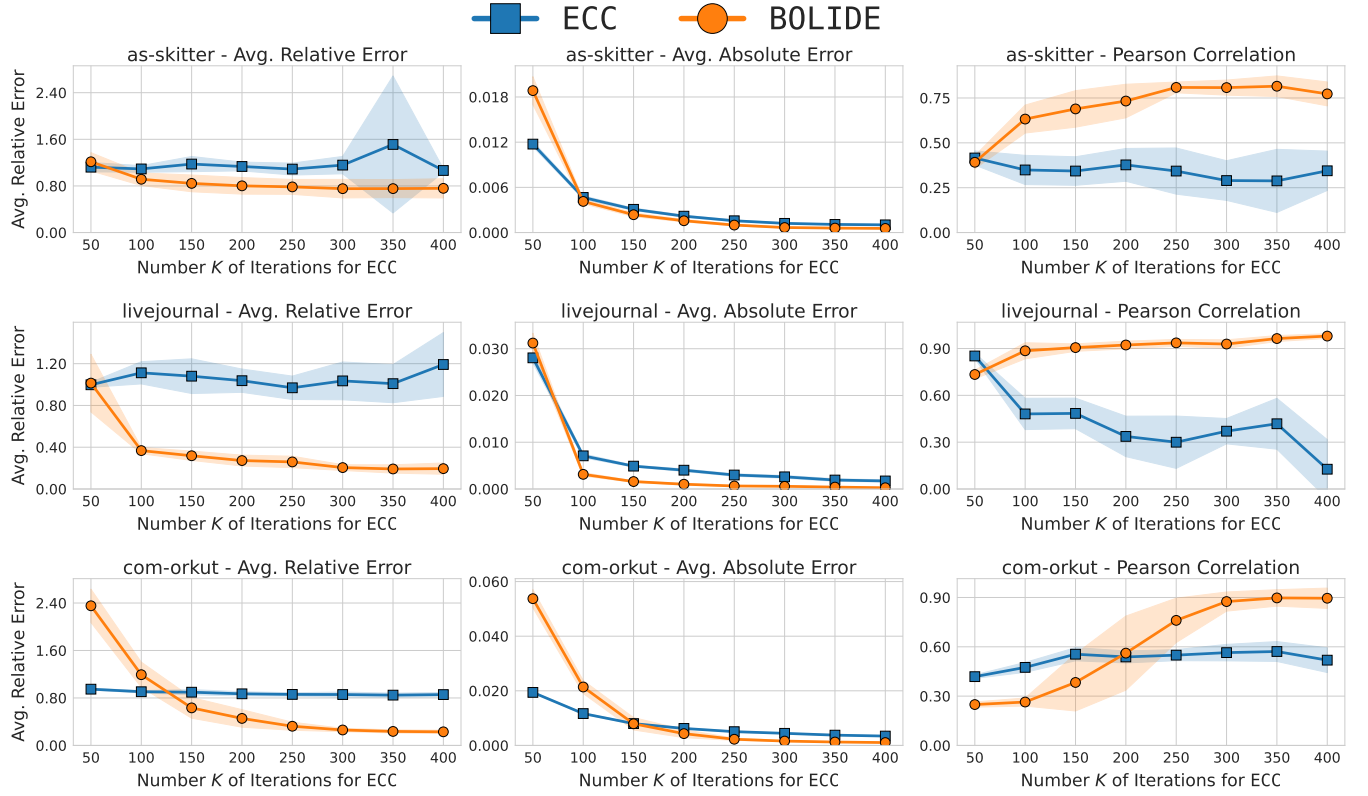


Figure 8: Comparison between ECC and BOLIDE. For each dataset (row), we report Average Relative Error, Average Absolute Error and Pearson Correlation coefficient over columns, by varying the number K of iterations for ECC. For each dataset and metric, the average and standard deviation over 10 independent repetitions are shown.